BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 264 · Operating-system functions

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does LMC LDA do?

Q2. What does the CPU control unit do?

Q3. Define a process.

Q4. Binary search reduces a sorted 32-element candidate set to one. How many successive halvings is this?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

process — Running/scheduled program instance

filesystem — Organised file storage

driver — Device interface software

spooling — Queued peripheral work

New learning

Understand — The OS mediates application requests and access to hardware.

Trace — Resource management schedules processes, allocates memory, controls files and drivers.

Explain — Permissions and isolation help keep processes and users from accessing unauthorised data.

Worked example

A pupil app requests a protected file. Explain the OS steps.

1. Predict first — The OS checks authorisation before returning content; a missing permission should prevent the read.

We Do · Practise with guidance

An app prints a document while a printer is busy. Explain the OS's role.

```
Application → print spool → driver → printer
```

Predict first

The print job enters a queue and is later sent via a compatible printer driver when the device is available.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State four operating-system responsibilities. [4]

Question 2

Describe a protected file read from application request to result. [6]

Question 3

Explain how an OS manages a browser, game and backup concurrently, including protection. [6]

Exit Ticket · Answer before checking

Applications must directly control all hardware to read files.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Loads ACC from a mailbox.

Q2. Coordinates and decodes execution.

Q3. An instance of a program being executed or scheduled.

Q4. 5: 32→16→8→4→2→1.

You Do pages · Independent questions

Q1. Process scheduling, memory management, file management and device management [1 each].

Q2. Request [1], identify user [1], authorisation [1], filesystem lookup [1], driver/device request [1], return or deny [1].

Q3. CPU scheduling [2], memory allocation/isolation [2], I/O management [1], permissions [1].

Exit Ticket · Model answer

False. The OS provides services and drivers.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: